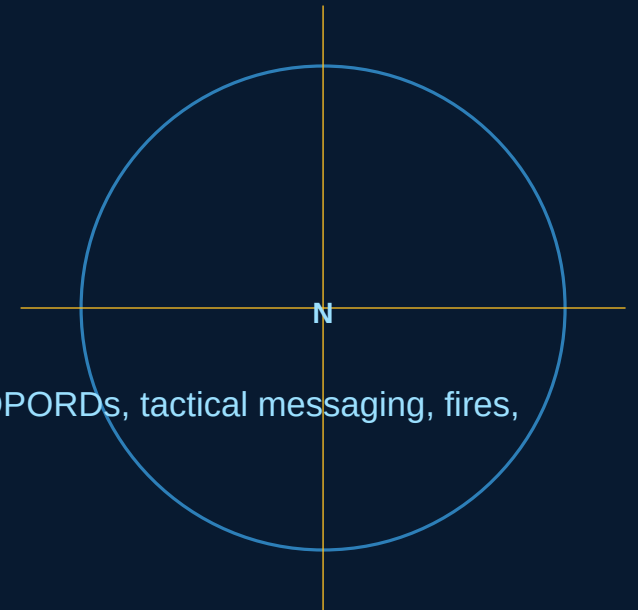


ARROW

Open situational awareness for the warfighter.

Faster OODA · Fewer screens · No black boxes.

A TAK-class soldier system — operator GPS, MIL-STD-2525, full NATO five-paragraph OPORDs, tactical messaging, fires, alerts and CoT bridge — on a stack you can read, run and own.



The problem.

Commercial SA tools solved this on paper a decade ago. In practice, small units still face four hard truths:

Fragmentation.

Map, chat, intel, the OPORD PDF and the contact report notebook never live in the same place. Every cross-domain hand-off is a screenshot or a verbal report over the net.

Closed ecosystems.

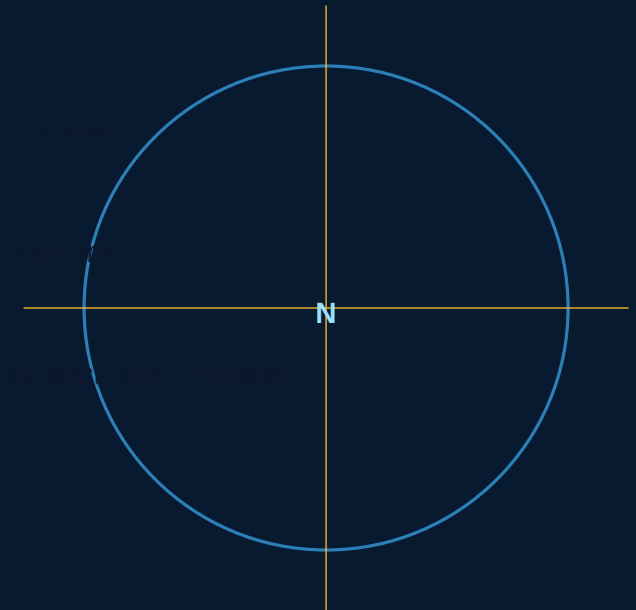
Vendor lock-in, opaque protocols, hardware you can't audit, license counts that don't survive a battalion rotation.

Slow OODA.

OPORDs live in PDFs, 9 Lines live on radios. The common operating picture is rarely common, rarely current.

Hard to interoperate.

ARPA speaks C4I, NATO speaks APP-11, partners speak what they were issued. Bridging them is a project, not a feature.



What ARROW is.

A modern soldier-system platform — open, inspectable, doctrinally honest.

Three tiers, one picture.

Backend tier — J2T with SQLAlchemy, Websocket publish

Task Battle Captain desktop — live map, OPORD editor, FDC

Android Impact-Capture client — OSINTnet, integrated, background GPS

Every domain is a FLUID, pluggable module: auth, tracking, alerts, messaging, reports, files, OPORDs, battle management, CoT, physics, streams.

By the numbers.

116

tactical objects / scenario

4

Ardennes COY plans

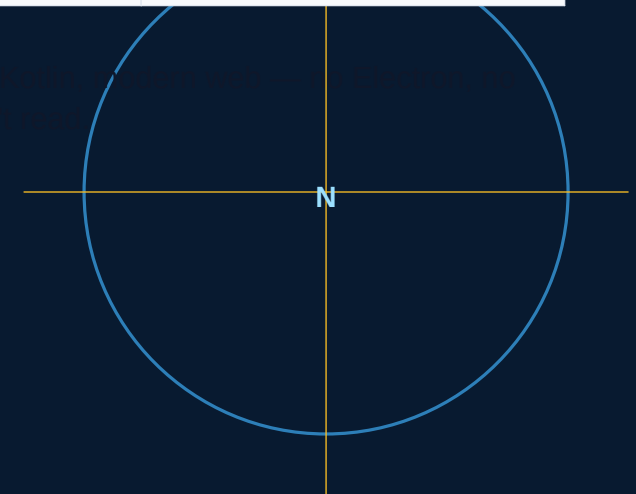
124

backend tests passing

24

live OPFOR units

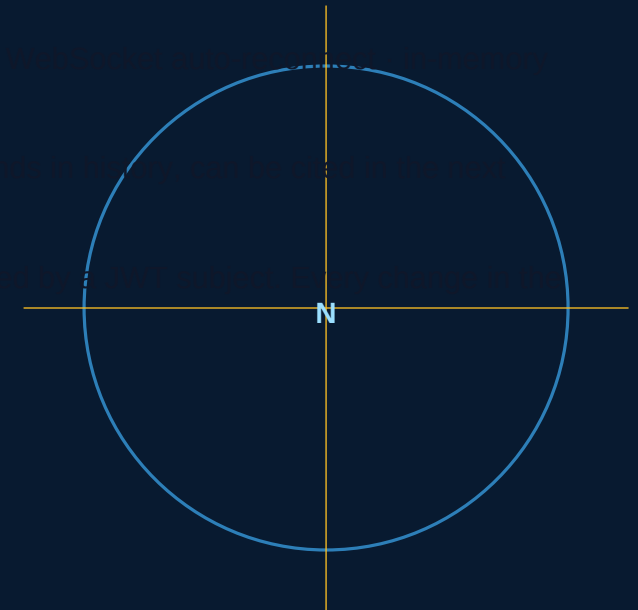
Built in Python 3.14, Kotlin, Kotlin web — Electron, no native UIs you can't reuse



Live tracking	Operational C2PS - intelligence sharing - real-time - 24/7 monitoring
MIL-STD-2525	Standardized friendly and enemy symbol systems - all tactical control graphics translated to common code - common language
OPORDs	All the paragraphs in JFTR's Operation Orders - never rendered map graphics with overlays - PDF export - available to operators - available on network
Tactical messaging	Real-time chat - voice - video - attachments - multi-personal - multi-channel - common language
Reports	Standardized - common language - common symbols - common data - common format
Fire missions	Standardized - common language - common symbols - common data - common format
Alerts	Standardized - common language - common symbols - common data - common format
Streams	Standard network - common C2PS - MIL-STD-2525 - common language
CoT bridge	Standard - common language - common symbols - common data - common format
Security	Standard network - common C2PS - MIL-STD-2525 - common language

What makes it different.

- Single source of truth. One in-process broadcast feeds every event to every client. Swap for Radio/NATC without touching callers.
- Open and auditable. Pure Python + Kotlin + standard web stack. Read the code, run the tests, deploy it. 12A with parsing.
- Doctrinally honest. OPORDs are real OPORDs. Symbols are real MIL-STD-2525. PACE plans are real PACE. We speak the operator's language, not a vendor's.
- Edge-tolerant. Cold tile cache + offline-core manifests + background tracking that survives down + WebSockets and local storage + memory token bucket when Redis is down.
- Cross-domain by design. A contact report drops a marker, fires an alert, broadcasts on chat, lands in history, can be used in the next OPORD — without any glue code.
- Built for ROE. Every action gated ADMIN / BATTLE / CAPTAIN / OPERATOR. Every event signed with MIT signed ID, a change in the audit log.

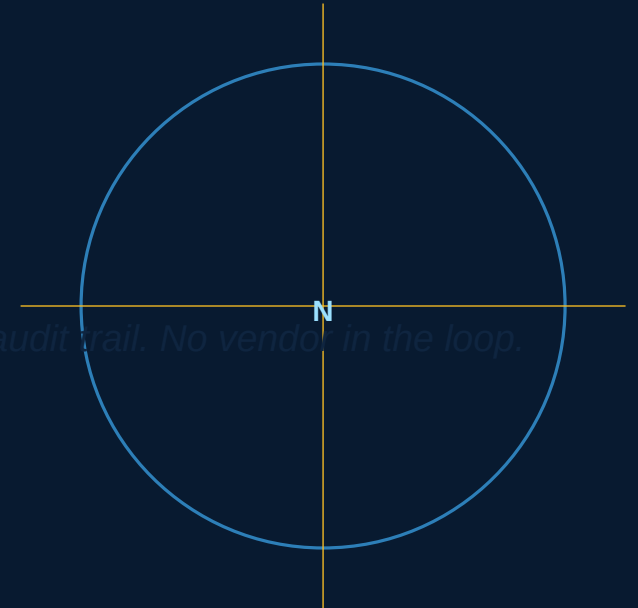


Why it matters — one screen, one fight.

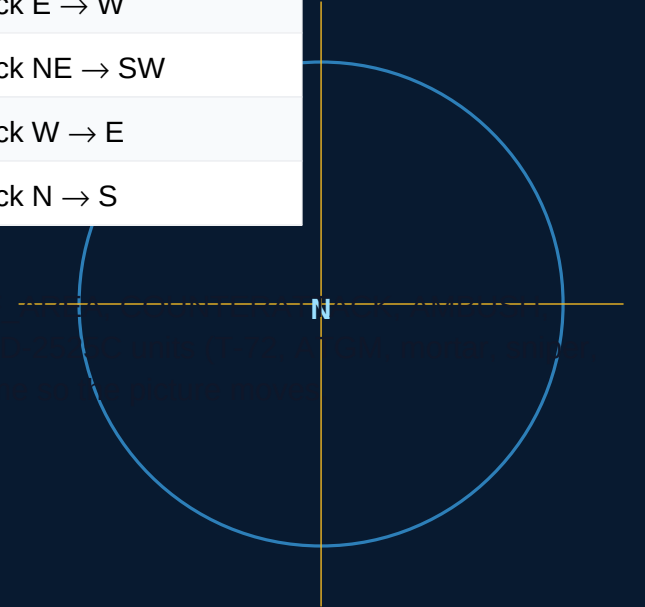
A platform leader with ARROW has one screen to:

- see what the people see,
- see what the enemy is doing,
- see the data that was brought her there,
- send the correct report,
- call for the right action,
- make the right call,
- ensure the team she got cleared,
- coordinate with the support team on call.

— from the same phone, on the same battery, on the same JWT, with the same audit trail. No vendor in the loop.
No license to expire. No data leaving her infrastructure.



Unit	Objective	Village	Axis
A CO	OBJ HAWK	Bastogne	attack E → W
B CO	OBJ EAGLE	Houffalize	attack NE → SW
C CO	OBJ FALCON	La Roche-en-Ardenne	attack W → E
D CO	OBJ KITE	Vielsalm	attack N → S



Architecture.

Backend — FastAPI

REST + GraphQL API (auto-generated schema provided). SQLAlchemy on SQLite (Postgres ready). One database instance has every domain even one channel tracking — tactical object — alert — chat — report — fire mission — report — stream — presence.

Web — Flask

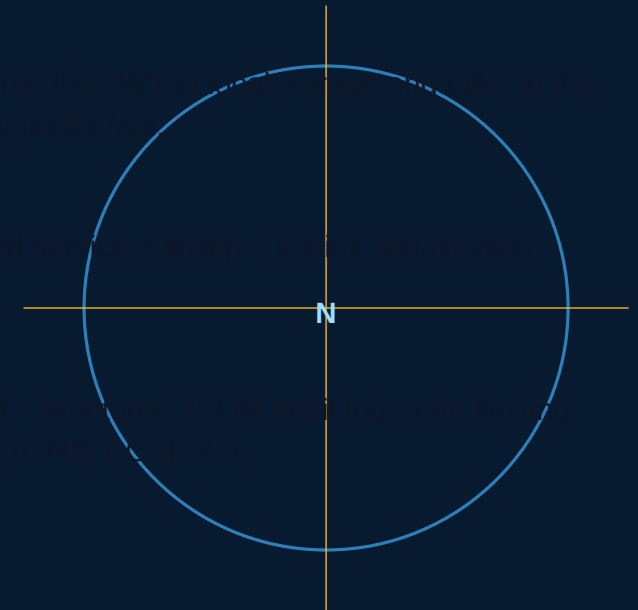
Operational dashboard. Each capability is a blueprint with its own routes and templates. Frontend stores the API keys and connects to the database backend directly. These serves only HTML/CSS/JS shells and a CSP hardened resource proxy under the hood.

Android — Jetpack Compose

Standalone Google project. Changelog map, material-equivalent rendering. Fixed Location foreground service. Context-aware sensor fusion. Composable native screens for every domain. Composition root in dsl/AppContainer.kt.

Security posture

API with auto-rotated secret — TOTP MFA — account expiry — AES-256-GCM photo encryption at rest — SHA-256 HMAC signing (not for integrity) — CSP / X-Frame-Options — Redis-backed token revocation (with in-memory fallback). Aligns to NIST SP 800-20.



Where it goes next.

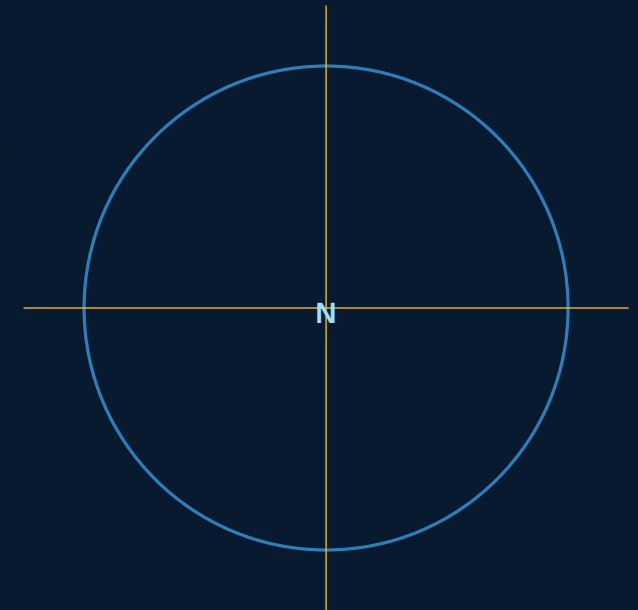
Multi-domain interoper. — JARCOT 2.0 — DoD GenPackage — NATO A2ADP-3 — OpenC2

Distributed AI. — Shephard —> automatic symbol recognition, chat —> automatic SITREP draft, sensor fusion across operator teams.

Federation. — Next of company TOCs grasping state —> disconnected ops, autonomous reconnection.

Coalition profile. — For partner data partners — RCE aware sharing — automatic redaction.

Hardware. — Robotic deployments on FLD, Samsung S20 Tactical, Boxer-class command vehicles.



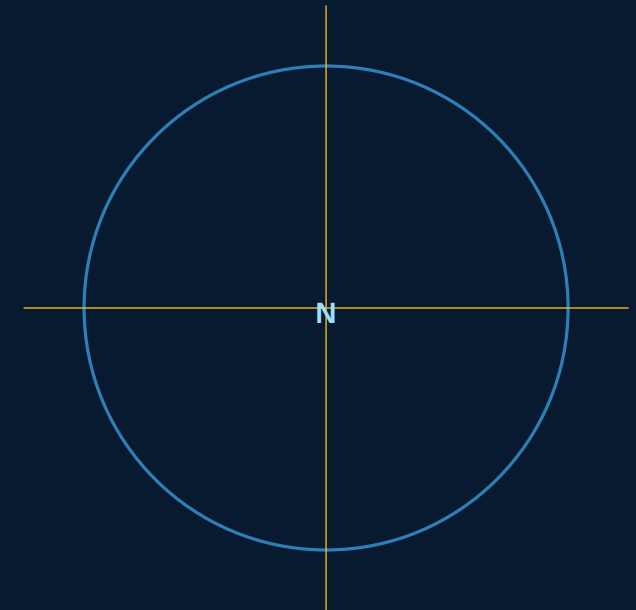
The ask.

Give us this — we give you a 24 platform your operators actually want to use, on infrastructure you own.

- A pilot battalion. 50 weeks, one rotation, real feedback.
- A partner SL. Harder the deployment story — Caddy, Kubernetes, FedRAMP.
- A coalition exercise. Prove CoT and APP 11 interoper in the wild.

ARROW — situational awareness, by warfighters, for warfighters.

Faster OODA. Fewer screens. No black boxes.



Thank you.

Let's get your platoons on one screen.

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